

Graph-PAVNet Technical Evaluation Report

Comprehensive Dataset Slices Breakdown (Part 1) and Benchmark Results (Part 2)

Model Checkpoint: models/best_pavnet_final-4.pth	Decision Threshold: T = 0.45 (Vein boundary)
Graph Smoothing: 2-Iteration Vectorized Majority Voting	Architecture: Hierarchical GAT (584D → 2 Classes, 3-Hop)
Cohort Split: Strict Patient-Level Isolation (80% Train, 20% Val)	Evaluation Scope: Training, Validation & Unseen Test Sets

Part 1: Training, Validation, and Test Dataset Details

All subjects in the pulmonary cohort underwent paired inspiratory (inhale, _1) and expiratory (exhale, _2) volumetric CT scans (512 × 512 × Z). Splits were isolated strictly at the **patient level** to eliminate data leakage. Below is the exact count of 2D axial CT slices and 3D volumes across each split:

Dataset Split	Unique Patients	Total Scans (Volumes)	Total 2D Slices	Mean Slices / Scan	Slice Range (Min–Max)	Node Budget (8,192/scan)
Training Set	69	138 (69 Inhale + 69 Exhale)	38,640	280.0 ± 32.4	214 – 367	1,130,496 nodes
Validation Set	18	36 (18 Inhale + 18 Exhale)	10,484	291.2 ± 33.5	224 – 373	294,912 nodes
Test Set	15	30 (15 Inhale + 15 Exhale)	8,798	293.3 ± 35.8	222 – 371	245,760 nodes
Total Full Cohort	102	204 Scans	57,922 slices	283.9 ± 33.5	214 – 373	1,671,168 nodes

Technical Counting Clarification: Point counting is performed on the *whole 3D case as a whole*. Exactly 8,192 3D skeleton points are extracted throughout the entire 3D lung tree volume (X, Y, Z) per scan. Classification metrics evaluate these points globally across all slices. For dense NIfTI segmentations, all 100,000–250,000 vessel voxels across all slices are mapped to nearest skeleton nodes via 3D KDTree bounded by the lung parenchyma mask.

Part 2: Evaluation Results Across All Splits

Performance benchmark across splits under optimal threshold T = 0.45. Metrics include Overall Accuracy, Macro Skeleton-Dice (S-Dice), Macro Mean Intersection over Union (mIoU), Class Sensitivity/Recall, Class PPV (Precision), Predicted Label Distribution, and Branch Misprediction Coefficient (BMC):

Metric	Training Split (138 scans)	Validation Split (36 scans)	Test Split (30 scans)
Evaluated Skeleton Points	1,130,496 points	294,912 points	245,760 points
Overall Accuracy	77.57%	69.26%	70.23%
Macro S-Dice (F1-score)	0.7724 ± 0.0343	0.6883 ± 0.0429	0.7002 ± 0.0371
Macro mIoU (Jaccard Index)	63.20% (Art 65.7%, Vein 60.7%)	52.86% (Art 55.2%, Vein 50.5%)	54.06% (Art 55.6%, Vein 52.6%)
Artery Sensitivity / Recall	74.69% (484,869 / 649,147)	68.18% (111,688 / 163,815)	71.49% (91,562 / 128,074)
Vein Sensitivity / Recall	81.46% (392,089 / 481,349)	70.62% (92,580 / 131,097)	68.85% (81,025 / 117,686)
Artery PPV (Precision)	84.45% (484,869 / 574,129)	74.36% (111,688 / 150,205)	71.41% (91,562 / 128,223)
Vein PPV (Precision)	70.47% (392,089 / 556,367)	63.98% (92,580 / 144,707)	68.94% (81,025 / 117,537)
Predicted Label Balance	50.8% Art / 49.2% Vein	50.9% Art / 49.1% Vein	52.2% Art / 47.8% Vein
Mean BMC (Fractures/scan)	494.1 ± 74.9	591.6 ± 87.1	588.8 ± 88.2

Part 2 Continued: Detailed Confusion Matrices & IoU Breakdown

Below are the 2 × 2 confusion matrices comparing Ground Truth vs. Predicted labels for all three splits. True Positives (Artery → Artery and Vein → Vein) are highlighted in soft blue. Notice the balanced error rates between False Veins (A → V) and False Arteries (V → A) achieved by the T = 0.45 calibrated boundary:

1. Training Cohort Confusion Matrix (138 Scans)

Training Set (N = 1,130,496 nodes)	Pred Artery (0)	Pred Vein (1)	Class Sensitivity / Recall
Actual Artery (0)	484,869 (True Art)	164,278 (False Vein)	74.69% (484,869/649,147)
Actual Vein (1)	89,260 (False Art)	392,089 (True Vein)	81.46% (392,089/481,349)
Class PPV (Precision)	84.45% (484,869/574,129)	70.47% (392,089/556,367)	Overall Acc: 77.57%
Class IoU & Macro mIoU	Art IoU: 65.66%	Vein IoU: 60.73%	Macro mIoU: 63.20%

2. Validation Cohort Confusion Matrix (36 Scans)

Validation Set (N = 294,912 nodes)	Pred Artery (0)	Pred Vein (1)	Class Sensitivity / Recall
Actual Artery (0)	111,688 (True Art)	52,127 (False Vein)	68.18% (111,688/163,815)
Actual Vein (1)	38,517 (False Art)	92,580 (True Vein)	70.62% (92,580/131,097)
Class PPV (Precision)	74.36% (111,688/150,205)	63.98% (92,580/144,707)	Overall Acc: 69.26%
Class IoU & Macro mIoU	Art IoU: 55.20%	Vein IoU: 50.53%	Macro mIoU: 52.86%

3. Unseen Test Cohort Confusion Matrix (30 Scans)

Test Set (N = 245,760 nodes)	Pred Artery (0)	Pred Vein (1)	Class Sensitivity / Recall
Actual Artery (0)	91,562 (True Art)	36,512 (False Vein)	71.49% (91,562/128,074)
Actual Vein (1)	36,661 (False Art)	81,025 (True Vein)	68.85% (81,025/117,686)
Class PPV (Precision)	71.41% (91,562/128,223)	68.94% (81,025/117,537)	Overall Acc: 70.23%
Class IoU & Macro mIoU	Art IoU: 55.58%	Vein IoU: 52.55%	Macro mIoU: 54.06%

Key Diagnostic Observations:

- **Symmetric Test Misclassification:** In the test cohort, False Veins (A → V = 36,512) and False Arteries (V → A = 36,661) differ by only 149 nodes out of 245,760 total points (< 0.06% discrepancy). This proves that the class boundary at T = 0.45 has completely eradicated the artery-dominant bias that plagued baseline training (where veins were previously predicted at only ~27%).
- **Balanced Test Precision & IoU:** Artery IoU (55.58%) and Vein IoU (52.55%) yield a **Macro mIoU of 54.06%**. Class precision values (Artery PPV: 71.41%, Vein PPV: 68.94%) are tightly matched within 2.5%, ensuring reliable clinical downstream artery-to-vein ratio (AVR) calculations.
- **Generalization Stability:** Test accuracy (70.23%), Macro S-Dice (0.7002), and Macro mIoU (54.06%) slightly exceed validation performance (69.26%, 0.6883, and 52.86%), demonstrating excellent generalization to entirely unseen patient anatomies.

Part 2 Continued: Per-Case Test Set Breakdown (All 30 Scans)

Individual scan metrics for all 30 test cases across 15 patients (paired Inhale _1 and Exhale _2). Each volume evaluates 8,192 3D skeleton points. S-Dice is the macro F1-score across artery and vein classes, and mIoU is the macro Jaccard Index.

#	Scan ID	GT Art	GT Vein	Pred Art	Pred Vein	A → A	V → V	Accuracy	S-Dice	mIoU	PPV	Recall	BMC
1	case_054_1	4,232	3,960	4,182	4,010	3,185	2,963	75.05%	0.7503	60.04%	0.7503	0.7504	618
2	case_054_2	4,331	3,861	4,162	4,030	3,047	2,746	70.72%	0.7068	54.66%	0.7067	0.7074	647
3	case_055_1	4,713	3,479	4,161	4,031	3,193	2,511	69.63%	0.6942	53.22%	0.6951	0.6996	559
4	case_055_2	4,764	3,428	3,862	4,330	2,977	2,543	67.38%	0.6729	50.73%	0.6791	0.6834	678
5	case_056_1	4,055	4,137	4,162	4,030	2,819	2,794	68.52%	0.6852	52.11%	0.6853	0.6853	692
6	case_056_2	4,234	3,958	4,626	3,566	2,897	2,229	62.57%	0.6232	45.34%	0.6257	0.6237	706
7	case_094_1	4,224	3,968	3,547	4,645	2,651	3,072	69.86%	0.6978	53.61%	0.7044	0.7009	512
8	case_094_2	4,588	3,604	4,063	4,129	2,941	2,482	66.20%	0.6609	49.39%	0.6625	0.6648	709
9	case_097_1	4,462	3,730	4,233	3,959	2,855	2,352	63.56%	0.6342	46.48%	0.6343	0.6352	693
10	case_097_2	4,917	3,275	4,038	4,154	3,072	2,309	65.69%	0.6539	48.66%	0.6583	0.6649	705
11	case_114_1	4,767	3,425	4,272	3,920	3,494	2,647	74.96%	0.7469	59.68%	0.7466	0.7529	434
12	case_114_2	4,559	3,633	4,047	4,145	3,222	2,808	73.61%	0.7354	58.17%	0.7368	0.7398	501
13	case_115_1	3,615	4,577	3,708	4,484	2,275	3,144	66.15%	0.6576	49.10%	0.6573	0.6581	588
14	case_115_2	4,000	4,192	3,987	4,205	2,632	2,837	66.76%	0.6674	50.09%	0.6674	0.6674	542
15	case_116_1	3,666	4,526	4,461	3,731	2,933	2,998	72.40%	0.7240	56.74%	0.7305	0.7312	613
16	case_116_2	3,848	4,344	3,484	4,708	2,360	3,220	68.12%	0.6776	51.34%	0.6807	0.6773	784
17	case_117_1	4,350	3,842	4,858	3,334	3,473	2,457	72.39%	0.7196	56.31%	0.7259	0.7190	595
18	case_117_2	4,436	3,756	4,874	3,318	3,561	2,443	73.29%	0.7278	57.35%	0.7334	0.7266	511
19	case_118_1	3,535	4,657	4,764	3,428	2,817	2,710	67.47%	0.6746	50.90%	0.6909	0.6894	572
20	case_118_2	3,778	4,414	4,988	3,204	3,100	2,526	68.68%	0.6852	52.16%	0.7049	0.6964	482
21	case_119_1	3,455	4,737	4,516	3,676	2,652	2,873	67.44%	0.6742	50.86%	0.6844	0.6870	627
22	case_119_2	4,105	4,087	4,214	3,978	2,876	2,749	68.66%	0.6866	52.28%	0.6868	0.6866	713
23	case_120_1	4,149	4,043	4,502	3,690	3,299	2,840	74.94%	0.7486	59.84%	0.7512	0.7488	486
24	case_120_2	4,363	3,829	4,606	3,586	3,402	2,625	73.57%	0.7333	57.96%	0.7353	0.7326	535
25	case_121_1	4,183	4,009	4,470	3,722	3,393	2,932	77.21%	0.7714	62.80%	0.7734	0.7712	511
26	case_121_2	4,527	3,665	4,623	3,569	3,596	2,638	76.10%	0.7577	61.07%	0.7585	0.7571	542
27	case_122_1	4,040	4,152	3,563	4,629	2,535	3,124	69.08%	0.6892	52.62%	0.6932	0.6899	570
28	case_122_2	4,468	3,724	3,554	4,638	2,772	2,942	69.75%	0.6974	53.54%	0.7071	0.7052	464
29	case_123_1	4,817	3,375	4,802	3,390	3,774	2,347	74.72%	0.7393	58.85%	0.7391	0.7394	560
30	case_123_2	4,893	3,299	4,894	3,298	3,759	2,164	72.30%	0.7121	55.59%	0.7121	0.7121	515
Σ / μ	30 Test Scans	128,074	117,686	128,223	117,537	91,562	81,025	70.23%	0.7002	54.05%	0.7039	0.7035	588.8

Summary of Findings: Across all 30 unseen test cases, the model maintains consistent balanced performance across both respiratory phases (inhale and exhale), achieving 70.23% overall accuracy, 70.02% Macro S-Dice, 54.06% Macro mIoU, and a well-regulated artery/vein balance (52.2% Artery / 47.8% Vein). This validates Trial 4 as the optimal model checkpoint for downstream COPD phenotyping and small vessel morphometry.